

Lecture 006

Classification

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Admin

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Material

Last time

- `tidymodels`
- Shrinkage methods
 - Ridge regression 
 - (The) lasso 
 - Elasticnet 

Today Classification methods

- Introduction to classification
- Linear probability models
- Logistic regression

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Upcoming

Readings *Today* ISL Ch. 4

Problem sets

- *Cross validation and shrinkage* Due next week.
- *Classification* After that...

Project Topic due

Classification

Classification

Intro

Regression problems seek to predict the number an outcome will take—integers (e.g., number of cats), reals (e.g., home/cat value), etc. [†]

[†] Maybe: Binary indicators...

Classification

Intro

Regression problems seek to predict the number an outcome will take—integers (e.g., number of cats), reals (e.g., home/cat value), etc. [†]

Classification problems instead seek to predict the category of an outcome

- **Binary outcomes**

success/failure; true/false; A or B; cat or *not cat*; etc.

- **Multi-class outcomes**

yes, no, *or maybe*; colors; letters; type of cat;^{††} etc.

This type of outcome is often called a *qualitative* or *categorical* response.

[†] Maybe: Binary indicators... ^{††} It turns out, all of machine learning is about cats.

Classification

Examples

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- Using life/criminal history (and demographics?):
Can we predict whether a defendant is **granted bail**?

Classification

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- Using life/criminal history (and demographics?):
Can we predict whether a defendant is **granted bail**?
- Based upon a set of symptoms and observations:
Can we predict a patient's **medical condition(s)**?

Classification

Examples

For the past few weeks, we've been immersed in regression problems.

It's probably helpful to mention a few **examples of classification problems**.

- Using life/criminal history (and demographics?):
Can we predict whether a defendant is **granted bail**?
- Based upon a set of symptoms and observations:
Can we predict a patient's **medical condition(s)**?
- From the pixels in an image:
Can we classify images as **bagel, puppy, or other**?

Classification

Approach

One can imagine two[†] related **approaches to classification**

1. Predict **which category** the outcome will take.
2. Estimate the **probability of each category** for the outcome.

[†] At least.

Classification

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1. Predict **which category** the outcome will take.
2. Estimate the **probability of each category** for the outcome.

That said, the general approach will

- Take a set of training observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Build a classifier $\hat{y}_o = f(x_o)$

all while balancing bias and variance.^{††}

[†] At least. ^{††} Sound familiar?

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A *Sometimes.*

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A *Sometimes. Other times:* No.

Q If everything is so similar, can't we use regression methods?

A *Sometimes. Other times:* No. Plus you still need new tools.

Classification

Why not regression?

Regression methods are not made to deal with **multiple categories**.

Ex. Consider three medical diagnoses: **stroke**, **overdose**, and **seizure**.

Regression needs a numeric outcome—how should we code our categories?

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$$Y = \begin{cases} 1 & \text{if } \text{stroke} \\ 2 & \text{if } \text{overdose} \\ 3 & \text{if } \text{seizure} \end{cases}$$

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Option 2

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Option 2

$$Y = \begin{cases} 1 & \text{if } \text{overdose} \\ 2 & \text{if } \text{stroke} \\ 3 & \text{if } \text{seizure} \end{cases}$$

Option 3

$$Y = \begin{cases} 1 & \text{if } \text{seizure} \\ 2 & \text{if } \text{stroke} \\ 3 & \text{if } \text{overdose} \end{cases}$$

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Option 2

$$Y = \begin{cases} 1 & \text{if overdose} \\ 2 & \text{if stroke} \\ 3 & \text{if seizure} \end{cases}$$

Option 3

$$Y = \begin{cases} 1 & \text{if seizure} \\ 2 & \text{if stroke} \\ 3 & \text{if overdose} \end{cases}$$

The categories' ordering is unclear—let alone the actual valuation.

The choice of ordering and valuation can affect predictions. 🙄

Classification

Why not regression?

As we've seen, **binary outcomes** are simpler.

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Why not regression?

As we've seen, **binary outcomes** are simpler.

Ex If we are only choosing between **stroke** and **overdose**

Option 1

$$Y = \begin{cases} 0 & \text{if } \text{stroke} \\ 1 & \text{if } \text{overdose} \end{cases}$$

and

Option 2

$$Y = \begin{cases} 0 & \text{if } \text{overdose} \\ 1 & \text{if } \text{stroke} \end{cases}$$

will provide the same results.

Classification

Why not regression?

In these **binary outcome** cases, we *can* apply linear regression.

These models are called **linear probability models** (LPMs).

The **predictions** from an LPM

1. estimate the conditional probability $y_i = 1$, i.e., $\Pr(y_o = 1 \mid x_o)$
2. are not restricted to being between 0 and 1[†]
3. provide an ordering—and a reasonable estimate of probability

[†] Some people get very worked up about this point.

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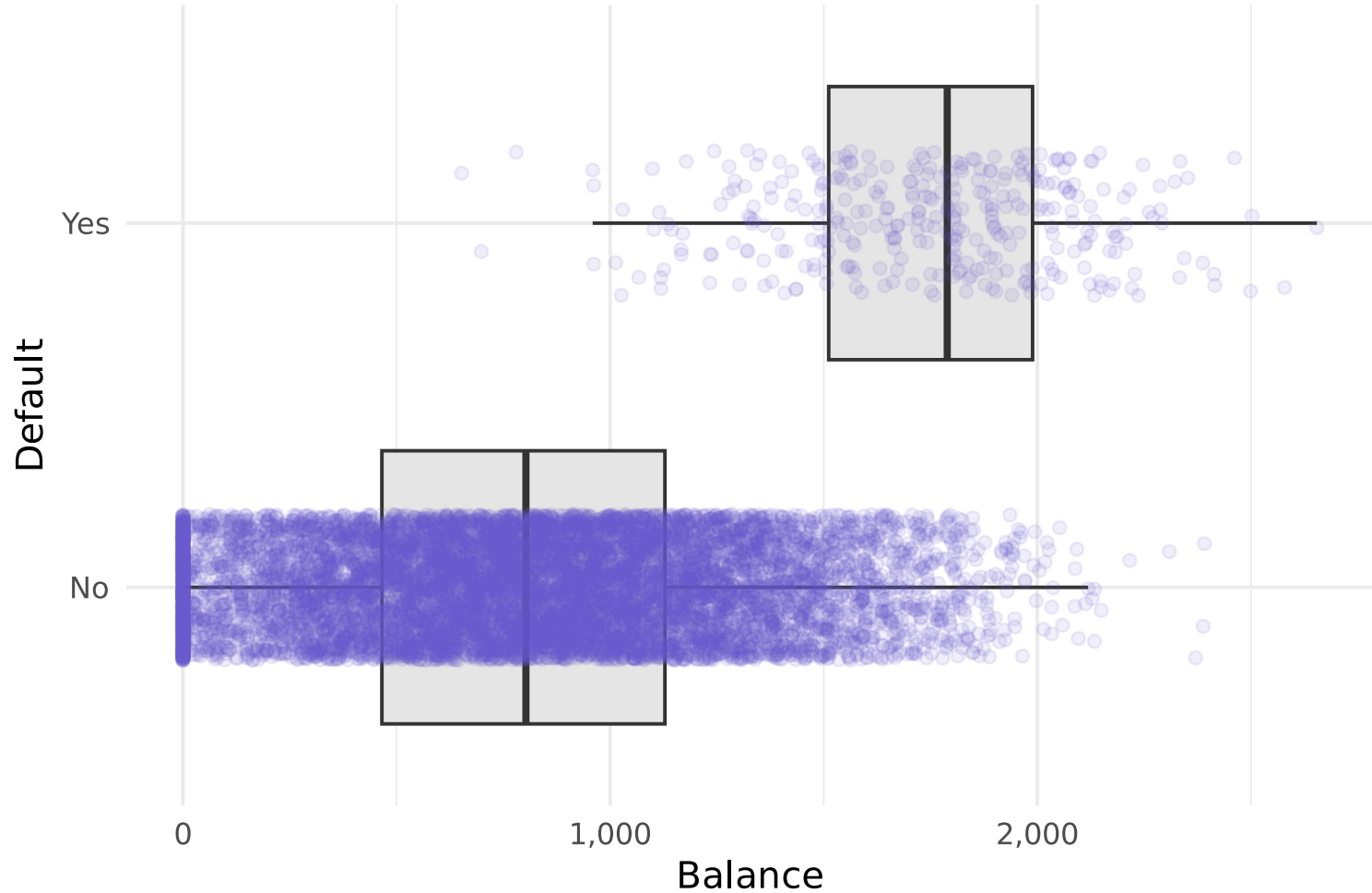
Other benefits: Coefficients are easily interpreted + we know how OLS works.

[†] Some people get very worked up about this point.

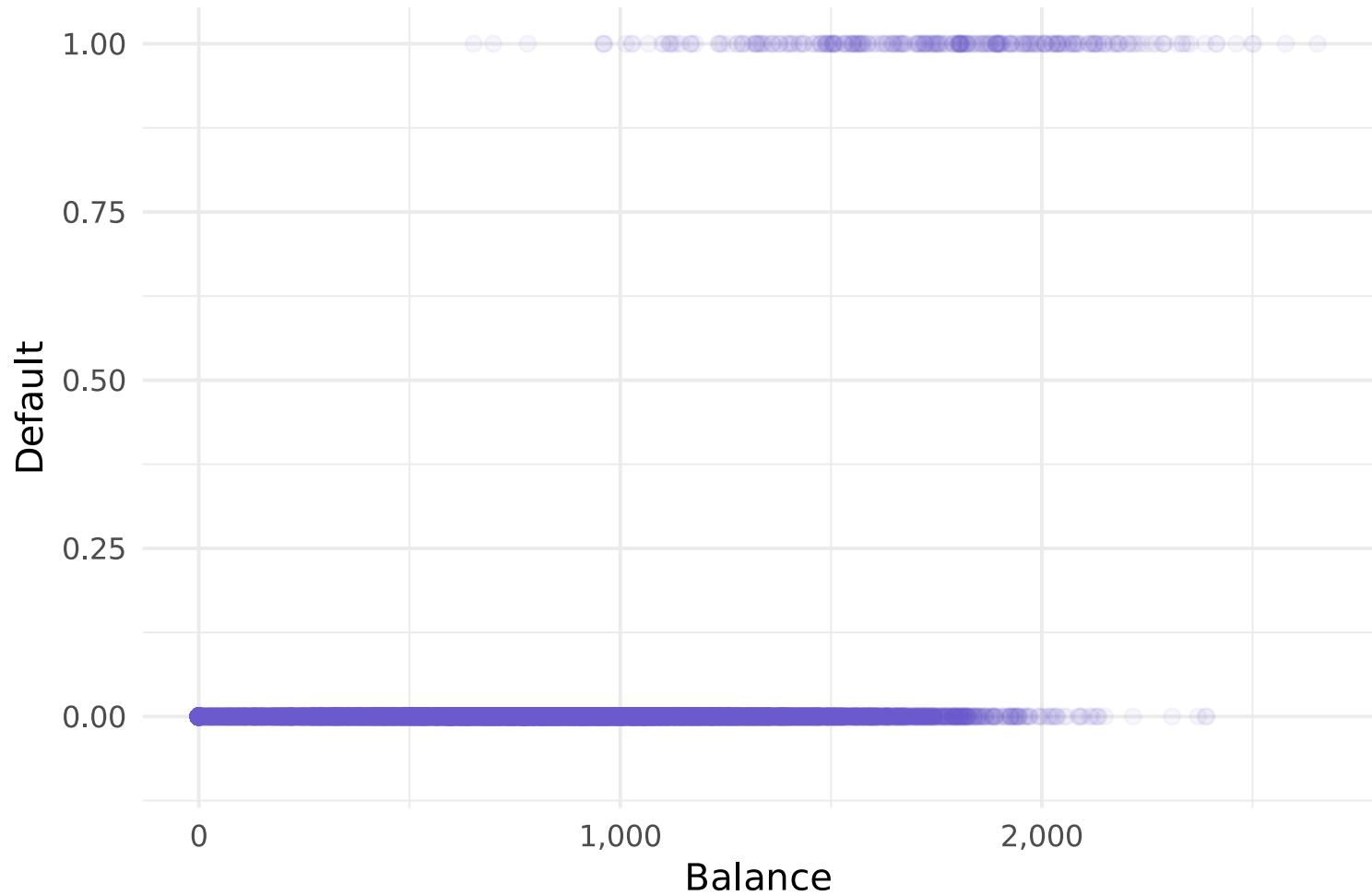
Let's consider an example: the `Default` dataset from `ISLR`

default	↕	student	↕	balance	↕	income	↕
No		No		939.10		45,519	
No		Yes		397.54		22,711	
Yes		No		1,511.61		53,507	
No		No		301.32		51,540	
No		No		878.45		29,562	
Yes		No		1,673.49		49,310	
No		No		310.13		37,697	
No		No		1,272.05		44,896	
No		No		887.20		41,641	
No		No		230.87		32,799	

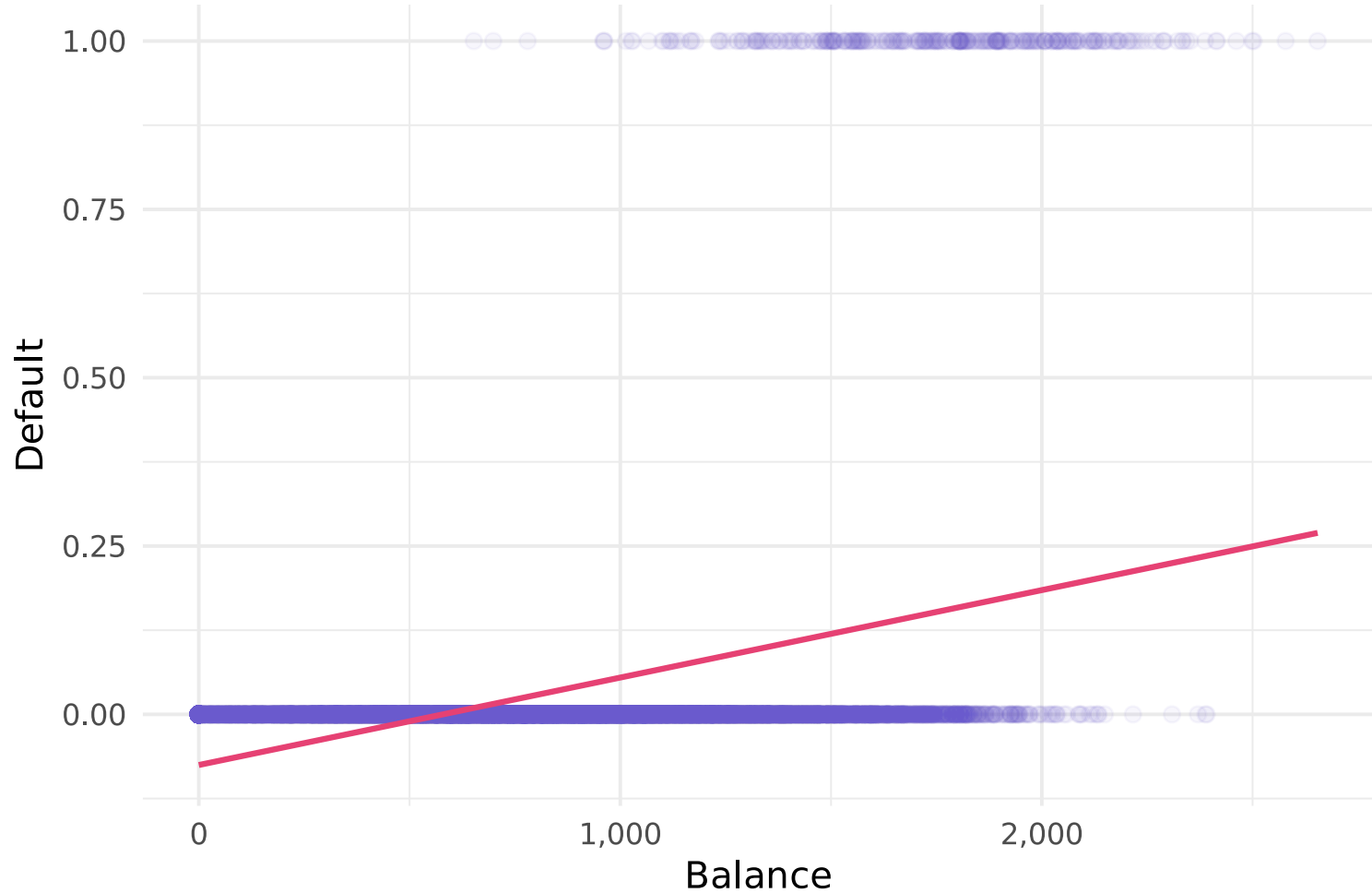
The data: The outcome, default, only takes two values (only 3.3% default).



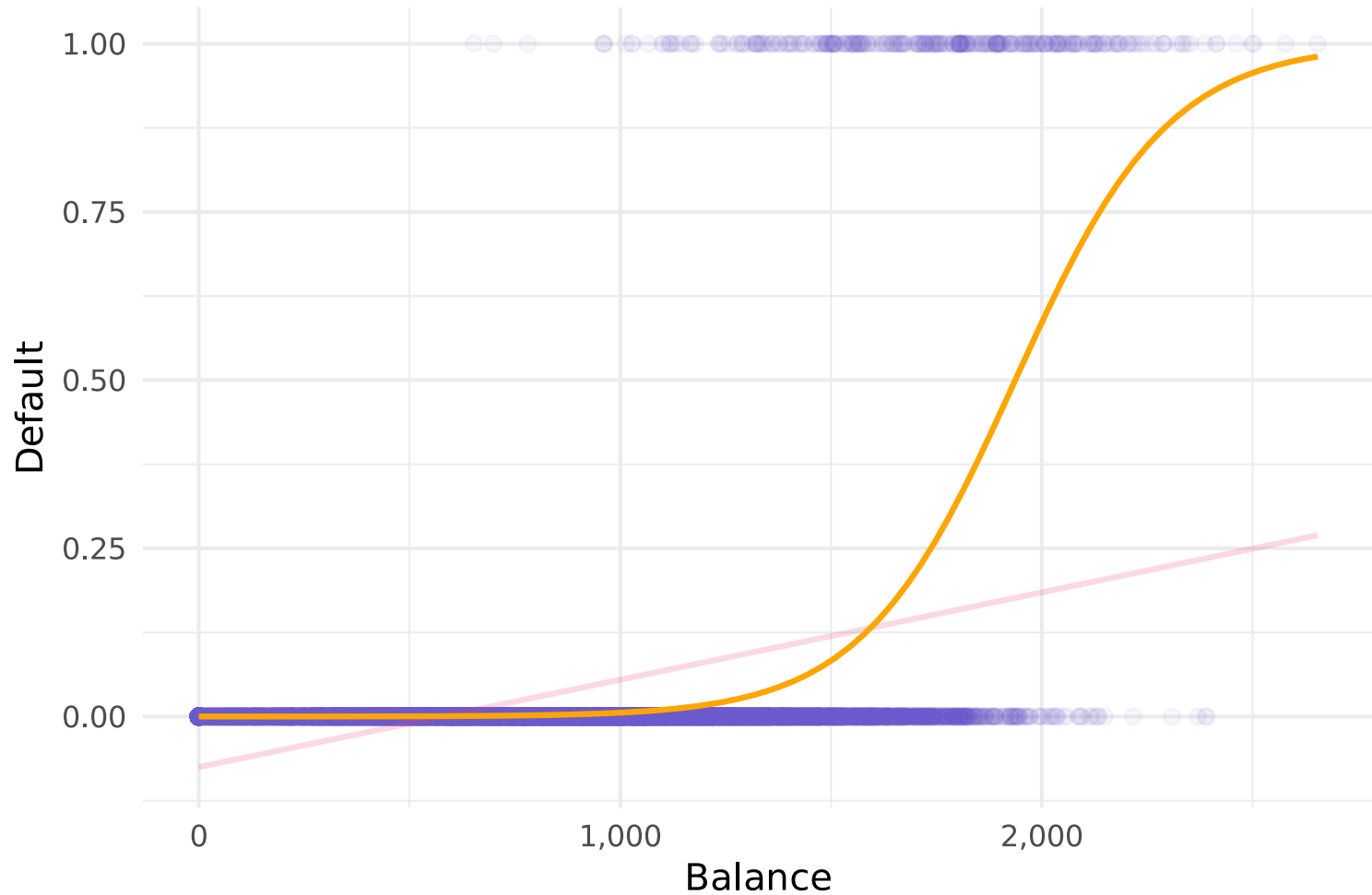
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The linear probability model struggles with prediction in this setting.



Logistic regression appears to offer an improvement.



So... what's logistic regression?

Logistic regression

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Intro

Logistic regression **models the probability** that our outcome Y belongs to a **specific category** (often whichever category we think of as `TRUE`).

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For example, we just saw a graph where

$$\Pr(\text{Default} = \text{Yes} | \text{Balance}) = p(\text{Balance})$$

we are modeling the probability of `default` as a function of `balance`.

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Logistic regression **models the probability** that our outcome Y belongs to a **specific category** (often whichever category we think of as `TRUE`).

For example, we just saw a graph where

$$\Pr(\text{Default} = \text{Yes} | \text{Balance}) = p(\text{Balance})$$

we are modeling the probability of `default` as a function of `balance`.

We use the **estimated probabilities** to **make predictions**, *e.g.*,

- if $p(\text{Balance}) \geq 0.5$, we could predict "Yes" for Default
- to be conservative, we could predict "Yes" if $p(\text{Balance}) \geq 0.1$

Logistic regression

What's *logistic*?

We want to model probability as a function of the predictors ($\beta_0 + \beta_1 X$).

Linear probability model

linear transform. of predictors

$$p(X) = \beta_0 + \beta_1 X$$

Logistic model

logistic transform. of predictors

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

What does this *logistic function* $\left(\frac{e^x}{1+e^x}\right)$ do?

1. takes some score (any real number) and maps it to a probability (0 to 1);
2. forces an S-shaped curve through the data (not linear).

Notice: We just connect probs. $p(X)$ to $\beta_0 + \beta_1 X$ with diff. transforms.

Logistic regression

What's *logistic*?

Aside: People also call the *logistic function* a **Sigmoid function** (S-shape).

This shape works well for modeling **natural and social phenomena**, e.g.,

- (constrained) population **growth/spread**,
- **learning** curves and **info/tech propagation**,
- **dose-response** and (crop) **yield** curves,

and is one of the fundamental elements of **neural networks**.

(Logistic regression itself is a very simple neural network.)

Logistic regression

What's *logistic*?

With a little math, you can show

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \implies \log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X$$

New definition: **log odds**[†] on the RHS and **linear predictors** on the LHS.

[†] The "log odds" is sometimes called "logit".

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New definition: **log odds**[†] on the RHS and **linear predictors** on the LHS.

1. **interpretation** of β_j is about **log odds**—not probability
2. **changes in probability** due to X depend on level of X ^{††}

[†] The "log odds" is sometimes called "logit". ^{††} It's nonlinear!

Logistic regression

Estimation

Before we can start predicting, we need to estimate the β_j s.

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \implies \log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X$$

We estimate logistic regression using **maximum likelihood estimation**.

Logistic regression

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We estimate logistic regression using **maximum likelihood estimation**.

Maximum likelihood estimation (MLE) searches for the parameters (β_j 's) that make our data "most likely" given the model we've written down.

Logistic regression

Maximum likelihood intuition

MLE searches for the β_j s that make our data "most likely" using our model.

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X$$

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2. odds = $\frac{p(X)}{1 - p(X)}$.

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$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X$$

1. β_j tells us how x_j affects the **log odds**

2. $\text{odds} = \frac{p(X)}{1 - p(X)}$. If $p(X) > 0.5$, then $\text{odds} > 1$ and **log odds** > 0 .

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So we want choose β_j such that

- **log odds** are above zero for observations where $y_i = 1$
- **log odds** even larger for areas of x_j where most i s have $y_i = 1$

Logistic regression

Formally: The likelihood function

Let's formalize the situation.

Let $\sigma(z) = \frac{e^z}{1 + e^z}$ be the logistic (sigmoid) function.

Then, for observation i ,

- *outcome* $Y_i \in \{0, 1\}$, where 1 indicates *success*,
- *dist* $Y_i \sim \text{Bernoulli}(p_i)$ with prob. of *success* p_i ,
- *prob* $p_i = \sigma(\beta_0 + \beta_1 x_{1,i} + \cdots + \beta_p x_{p,i}) = \sigma(x'_i \beta)$;

A single **Bernoulli** experiment has **probability mass function** (PMF)

$$P(Y_i = y | X_i = x_i) = p_i^y (1 - p_i)^{1-y} = \sigma(x'_i \beta)^y [1 - \sigma(x'_i \beta)]^{1-y}$$

Logistic regression

Formally: The likelihood function

Define The **likelihood function** $\mathcal{L}(\beta)$ is the **joint probability** of observing our data as a function of the parameters (β) .

Recall The **joint probability** of independent events is the *product* of their **individual probabilities**.

So the **likelihood function** is the *product* of individual probabilities, given β

$$\begin{aligned}\mathcal{L}(\beta) &= \prod_{i=1}^n P(Y_i = y_i | X_i = x_i) \\ &= \prod_{i=1}^n \sigma(x_i' \beta)^{y_i} [1 - \sigma(x_i' \beta)]^{1-y_i}\end{aligned}$$

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Notice: Different choices of β lead to different **likelihoods** (we want to max).

Logistic regression

Formally: The log likelihood function

From the *likelihood function*

$$\mathcal{L}(\beta) = \prod_{i=1}^n \sigma(x'_i \beta)^{y_i} [1 - \sigma(x'_i \beta)]^{1-y_i}$$

we can get the **log-likelihood function**

$$\ell(\beta) = \sum_{i=1}^n (y_i \log[\sigma(x'_i \beta)] + (1 - y_i) \log[1 - \sigma(x'_i \beta)])$$

which is often nicer to work with and still achieves the same maximum.

Logistic regression

Formally: Maximum likelihood estimation

We **estimate** logistic reg. by choosing the β s that **max.** (log) **likelihood**

$$\begin{aligned}\ell(\beta) &= \sum_{i=1}^n (y_i \log[\sigma(x'_i \beta)] + (1 - y_i) \log[1 - \sigma(x'_i \beta)]) \\ &= \sum_{i:y_i=1} \log[\sigma(x'_i \beta)] + \sum_{i:y_i=0} \log[1 - \sigma(x'_i \beta)] \\ &= \sum_{i:y_i=1} \log[p(x_i)] + \sum_{i:y_i=0} \log[1 - p(x_i)]\end{aligned}$$

Intuition Max. (log) likelihood function by choosing β s that

- make $p(x_i)$ large for individuals with $y_i = 1$,
- make $p(x_i)$ small for individuals with $y_i = 0$.

Logistic regression

Formally: Finding the max

MLE wants to find the vector β that maximizes $\ell(\beta)$.

Idea 1 Search blindly over (infinite) parameter space.

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(And check second-order conditions.)

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(And check second-order conditions.)

Problem No closed-form solution for logistic regression.

That said, partial derivatives will still be helpful...

$$\frac{\ell(\beta)}{\beta_j} = \sum_{i=1}^n [y_i - \sigma(x'_i \beta)] x_{j,i}$$

Logistic regression

Formally: Numerical optimization

Idea 1 + Idea 2 = Idea 3 Numerical optimization algorithms.

Gradient ascent uses derivatives (gradients) to iteratively improve guesses.

- start with a naïve guess of parameters,
- evaluate the gradient at that guess, (*where is steepest ascent?*)
- update your guess in the direction of the gradient. (*move up the hill*)
- repeat. (*keep hiking until you reach the peak*)

$$\beta_j^{(\text{new})} = \beta_j^{(\text{old})} + \eta \frac{\partial \ell(\beta)}{\partial \beta_j}(\beta^{(\text{old})})$$
$$\implies \beta_j^{(\text{new})} = \beta_j^{(\text{old})} + \eta \sum_{i=1}^n [y_i - \sigma(x_i' \beta^{(\text{old})})] x_{j,i}$$

Logistic regression

Maximum likelihood estimation

A few notes

1. η is the **learning rate** (step size);
too big and you might overshoot, too small and it takes forever.
2. Gradient ascent is one of many numerical optimization algorithms.
Others include **Newton-Raphson** and **BFGS**.
3. As with other estimators, you'll use **existing software** in practice.
4. *Search algorithms* are central to many models, e.g., **neural networks**.

Logistic regression

In R

In R, you can run logistic regression using the `glm()` function.

Also: `logistic_reg()` in the `tidymodels` galaxy (with the `"glm"` engine).

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Aside: Related to `lm`, `glm` stands for *generalized* (linear model).

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Aside: Related to `lm`, `glm` stands for *generalized* (linear model).

"Generalized" essentially means that we're applying some transformation to $\beta_0 + \beta_1 X$ like logistic regression applies the logistic function.

More generally:

$$\mathbf{y} = g^{-1}(\mathbf{X}\boldsymbol{\beta}) \iff g(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$$

Logistic regression

In R

In R, you can run logistic regression using the `glm()` function.

Key arguments (very similar to `lm()`)

- specify a formula,[†] e.g., `y ~ .` or `y ~ x + I(x^2)`
- define `family = "binomial"` (so R knows to run logistic regression)
- give the function some `data`

[†] Notice that we're back in the world of needing to select a model...

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```
est_logistic = glm(  
  i_default ~ balance,  
  family = "binomial",  
  data = default_df  
)
```

[†] Notice that we're back in the world of needing to select a model...

```
est_logistic ▷ summary()
```

```
#>
#> Call:
#> glm(formula = i_default ~ balance, family = "binomial", data = default_df)
#>
#> Coefficients:
#>              Estimate Std. Error z value Pr(>|z|)
#> (Intercept) -1.065e+01  3.612e-01  -29.49   <2e-16 ***
#> balance      5.499e-03  2.204e-04   24.95   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> (Dispersion parameter for binomial family taken to be 1)
#>
#>      Null deviance: 2920.6  on 9999  degrees of freedom
#> Residual deviance: 1596.5  on 9998  degrees of freedom
#> AIC: 1600.5
#>
#> Number of Fisher Scoring iterations: 8
```

Logistic regression

Estimates and predictions

Thus, our estimates are $\hat{\beta}_0 \approx -10.65$ and $\hat{\beta}_1 \approx 0.0055$.

Remember: These coefficients are for the **log odds**.

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Remember: These coefficients are for the **log odds**.

If we want **to make predictions** for y_i (whether or not i defaults), then we first must **estimate the probability** $p(\text{Balance})$

$$\hat{p}(\text{Balance}) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 \text{Balance}}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 \text{Balance}}} \approx \frac{e^{-10.65 + 0.0055 \cdot \text{Balance}}}{1 + e^{-10.65 + 0.0055 \cdot \text{Balance}}}$$

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Thus, our estimates are $\hat{\beta}_0 \approx -10.65$ and $\hat{\beta}_1 \approx 0.0055$.

Remember: These coefficients are for the **log odds**.

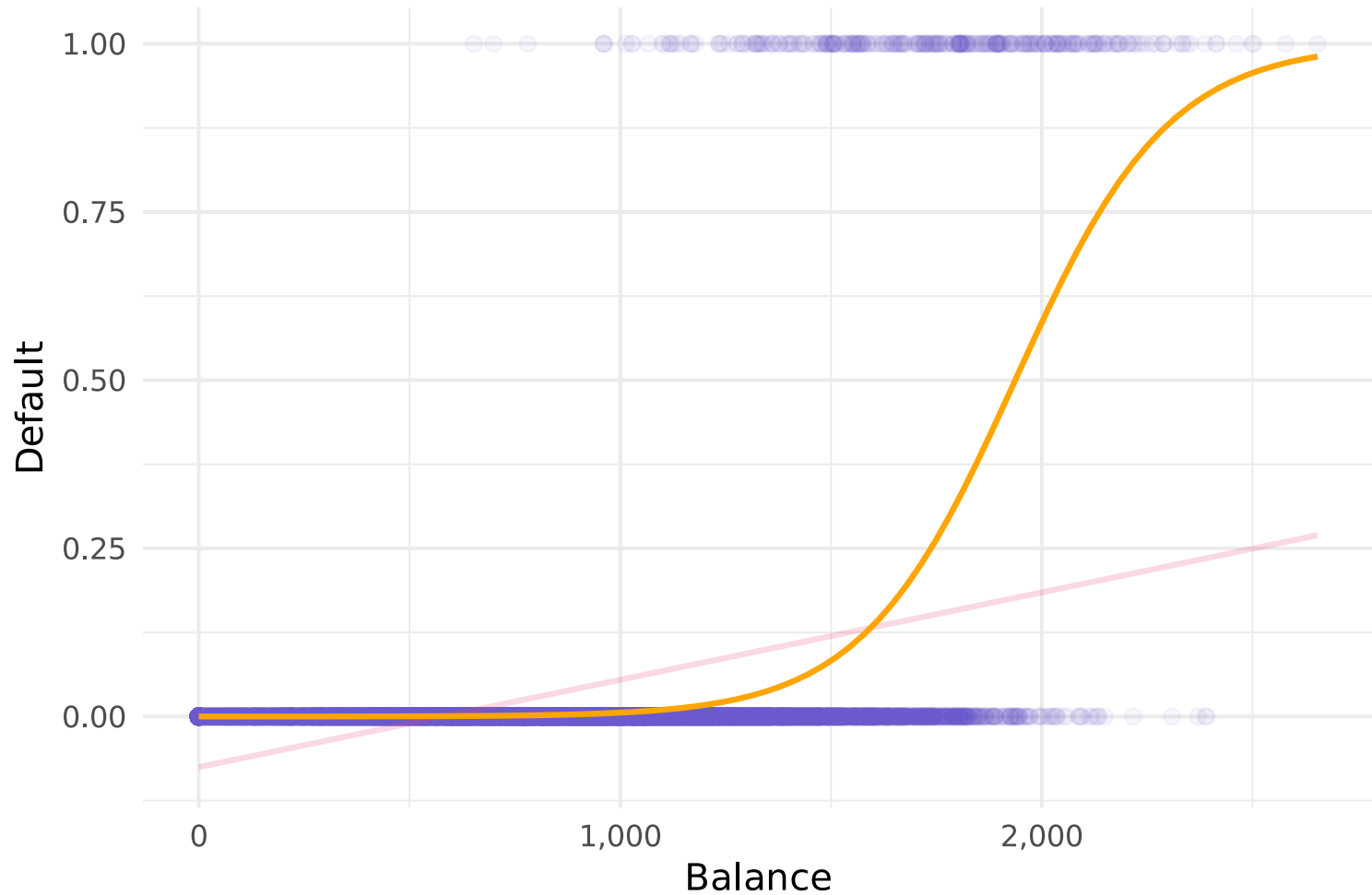
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$$\hat{p}(\text{Balance}) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 \text{Balance}}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 \text{Balance}}} \approx \frac{e^{-10.65 + 0.0055 \cdot \text{Balance}}}{1 + e^{-10.65 + 0.0055 \cdot \text{Balance}}}$$

- If $\text{Balance} = 0$, we then estimate $\hat{p} \approx 0.000024$
- If $\text{Balance} = 2,000$, we then estimate $\hat{p} \approx 0.586$
- If $\text{Balance} = 3,000$, we then estimate $\hat{p} \approx 0.997$ [†]

[†] You get a sense of the nonlinearity of the predictors' effects.

Logistic regression's predictions of $p(\text{Balance})$



Note: Everything we've done so far extends to models with many predictors.

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Old news: You can use `predict()` to get predictions out of `glm` objects.

New and important: `predict()` produces multiple `types` of predictions

1. `type = "response"` predicts *on the scale of the response variable* for logistic regression, this means **predicted probabilities** (0 to 1)
2. `type = "link"` predicts *on the scale of the linear predictors* for logistic regression, this means **predicted log odds** ($-\infty$ to ∞)

Beware: The default is `type = "link"`, which you may not want.

Logistic regression

Prediction

Putting it all together, we can get (estimated) probabilities $\hat{p}(X)$

```
# Predictions on scale of response (outcome) variable  
p_hat = predict(est_logistic, type = "response")
```

which we can use to make predictions on y

```
# Predict '1' if p_hat is greater or equal to 0.5  
y_hat = as.numeric(p_hat ≥ 0.5)
```

So how did we do?

Assessment

Assessment

How did we do?

We guessed 97.25% of the observations correctly.

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[†] This idea is called the *null classifier*.

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Q How can we more formally assess our model's performance?

A All roads lead to the **confusion matrix**.

[†] This idea is called the *null classifier*.

Assessment

The confusion matrix

The **confusion matrix** is us a convenient way to display **correct** and **incorrect** predictions for each class of our outcome.

		Truth	
		No	Yes
Prediction	No	True Negative (TN)	False Negative (FN)
	Yes	False Positive (FP)	True Positive (TP)

Assessment

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The **accuracy** of a method is the share of **correct** predictions, *i.e.*,

$$\text{Accuracy} = (\text{TN} + \text{TP}) / (\text{TN} + \text{TP} + \text{FN} + \text{FP})$$

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This matrix also helps display many other measures of assessment.

Assessment

The confusion matrix

Sensitivity: the share of positive outcomes $Y = 1$ that we correctly predict.

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$$

		Truth	
		No	Yes
Prediction	No	True Negative (TN)	False Negative (FN)
	Yes	False Positive (FP)	True Positive (TP)

Sensitivity is also called **recall** and the **true-positive rate**.

One minus sensitivity is the **type-II error rate**.

Assessment

The confusion matrix

Specificity: the share of neg. outcomes ($Y = 0$) that we correctly predict.

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

		Truth	
		No	Yes
Prediction	No	True Negative (TN)	False Negative (FN)
	Yes	False Positive (FP)	True Positive (TP)

One minus specificity is the **false-positive rate** or **type-I error rate**.

Assessment

The confusion matrix

Precision: the share of predicted positives ($\hat{Y} = 1$) that are correct.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

		Truth	
		No	Yes
Prediction	No	True Negative (TN)	False Negative (FN)
	Yes	False Positive (FP)	True Positive (TP)

Assessment

The confusion matrix

Negative prediction value:

the share of predicted negatives ($\hat{Y} = 0$) that are correct.

$$\text{NPV} = \text{TN} / (\text{TN} + \text{FN})$$

		Truth	
		No	Yes
Prediction	No	True Negative (TN)	False Negative (FN)
	Yes	False Positive (FP)	True Positive (TP)

Note: NPV is not commonly used.

Assessment

Which assessment?

Q So *which* criterion should we use?

Assessment

Which assessment?

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- Are true positives more valuable than true negatives?

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- Do you want to have high confidence in predicted positives?

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Precision is your friend

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- Do you want to have high confidence in predicted positives?
Precision is your friend
- Are all errors equal?

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Precision is your friend
- Are all errors equal?
Accuracy is perfect.

Assessment

Which assessment?

Q So *which* criterion should we use?

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- Are true positives more valuable than true negatives?
Sensitivity will be key.
- Do you want to have high confidence in predicted positives?
Precision is your friend
- Are all errors equal?
Accuracy is perfect.

There's a lot more, e.g., the **F₁ score** combines precision and sensitivity.

Assessment

Confusion in R

`conf_mat()` from `yardstick` (`tidymodels`) calculates the confusion matrix.

- `data`: a dataset (`factor` variables) of true values and predictions
- `truth`: the name of the column (in `data`) of the truth values
- `estimate`: the name of the column (in `data`) of our predictions

Assessment

Confusion in R

`conf_mat()` from `yardstick` (`tidymodels`) calculates the confusion matrix.

- `data`: a dataset (`factor` variables) of true values and predictions
- `truth`: the name of the column (in `data`) of the truth values
- `estimate`: the name of the column (in `data`) of our predictions

```
cm_logistic = conf_mat(  
  # Create a dataset of truth and predictions  
  data = tibble(  
    y_hat = y_hat %>% as.factor(),  
    y = default_df$i_default %>% as.factor()  
  ),  
  truth = y, estimate = y_hat  
)
```

Assessment

Confusion in R

`conf_mat()` from `yardstick` (`tidymodels`) calculates the confusion matrix.

- `data`: a dataset (`factor` variables) of true values and predictions
- `truth`: the name of the column (in `data`) of the truth values
- `estimate`: the name of the column (in `data`) of our predictions

```
#>           Truth
#> Prediction    0    1
#>           0 9625 233
#>           1   42 100
```

Assessment

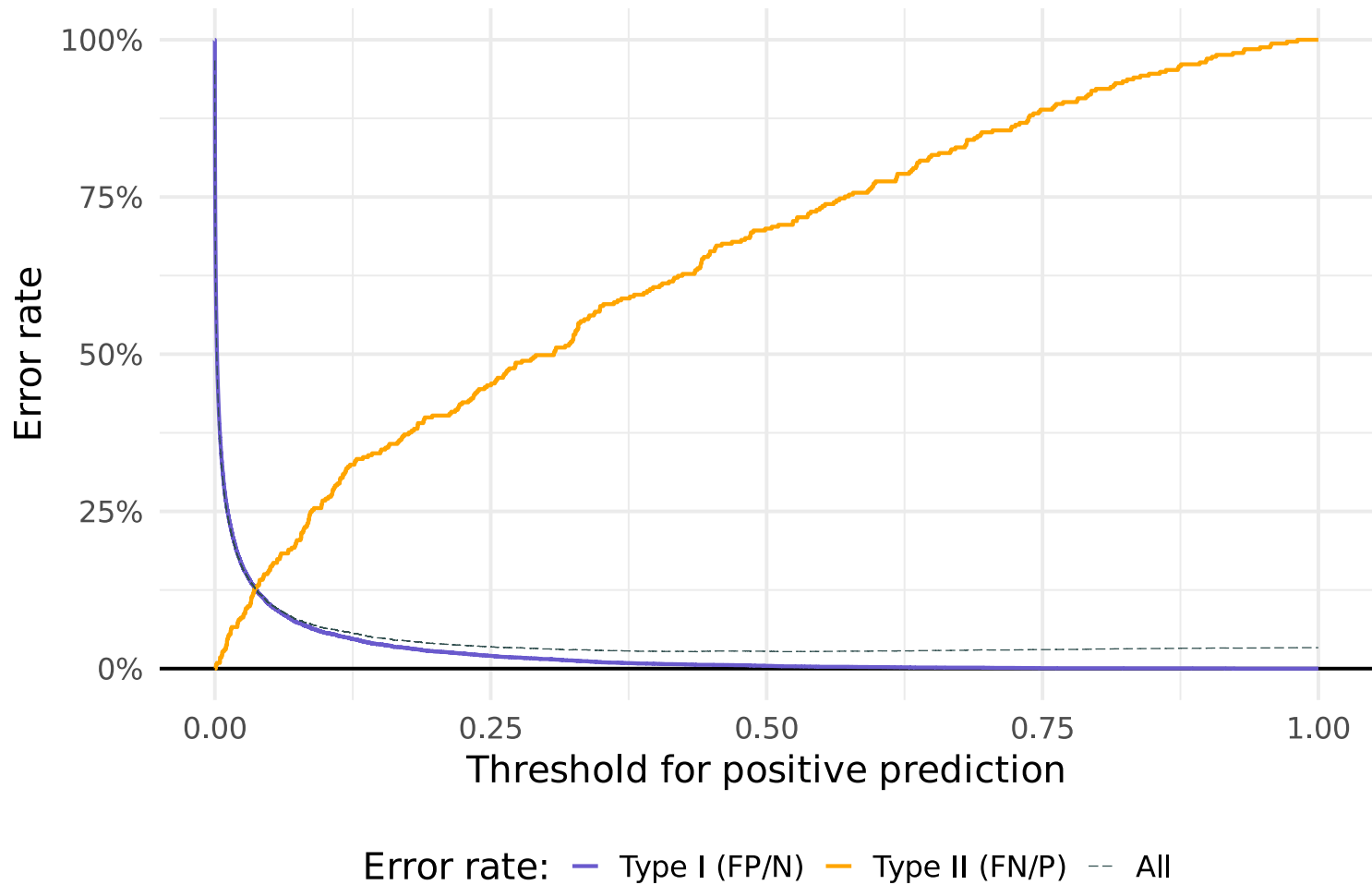
Thresholds

Your setting also dictates the "optimal" threshold that moves a prediction from one class (e.g., Default = No) to another class (Default = Yes).

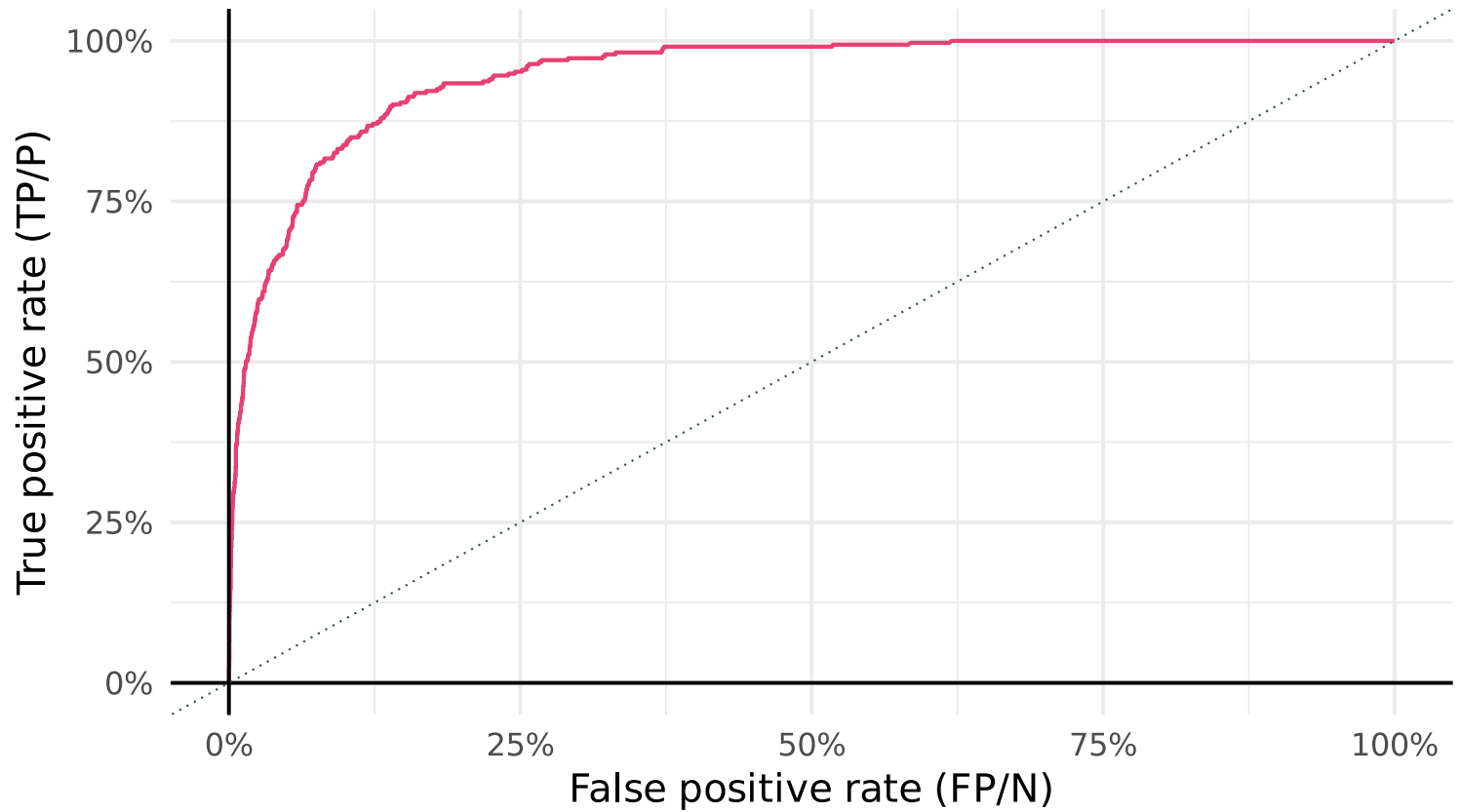
The Bayes classifier suggests a probability threshold of 0.5.

The Bayes classifier can't be beat in terms of *accuracy*, but if you have goals other than accuracy, you should consider other thresholds.

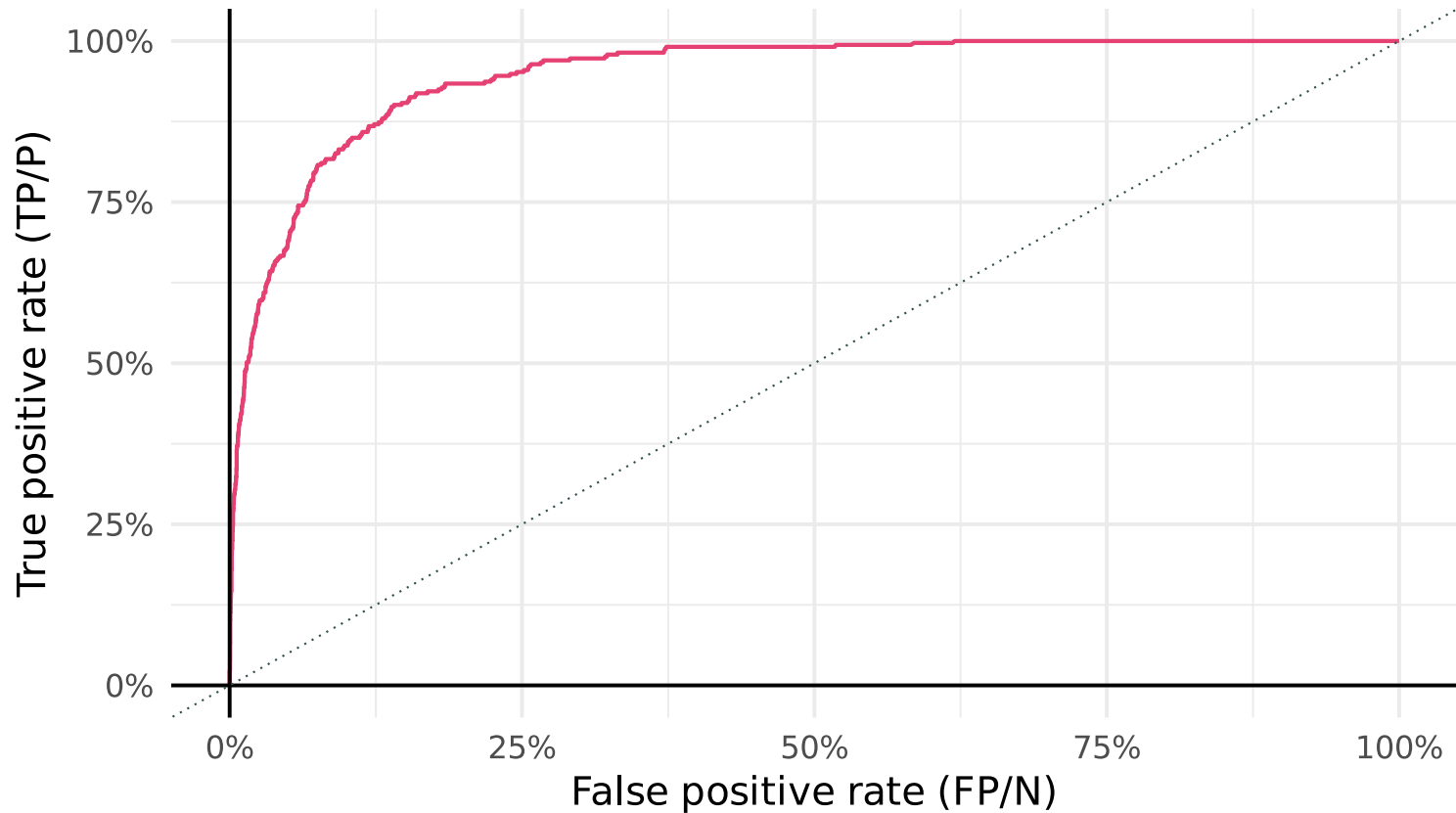
As we vary the threshold, our error rates (types I, II, and **overall**) change.



The **ROC curve** plots the true- (TP/P) and the false-positive rates (FP/N).

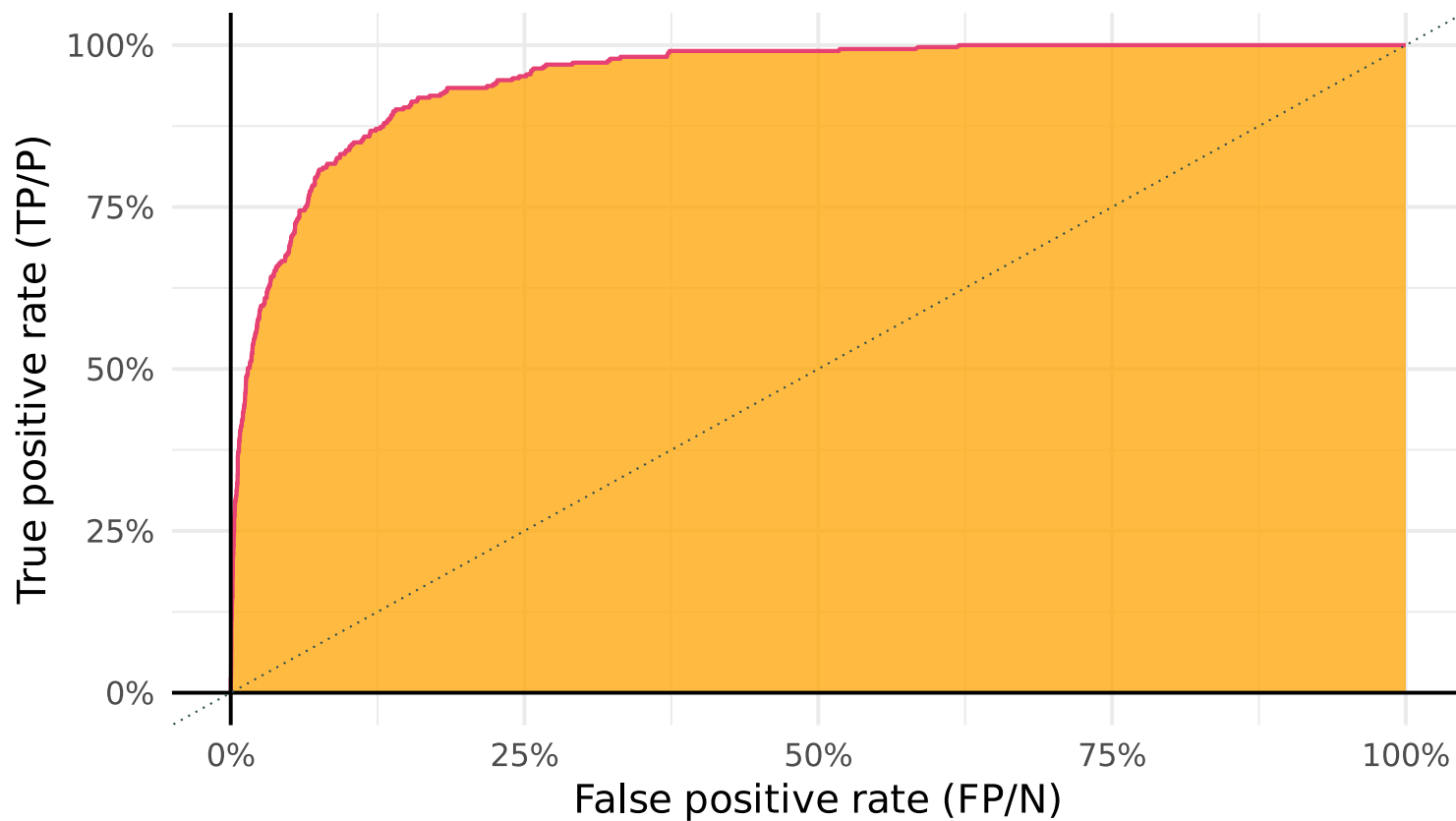


The **ROC curve** plots the true- (TP/P) and the false-positive rates (FP/N).

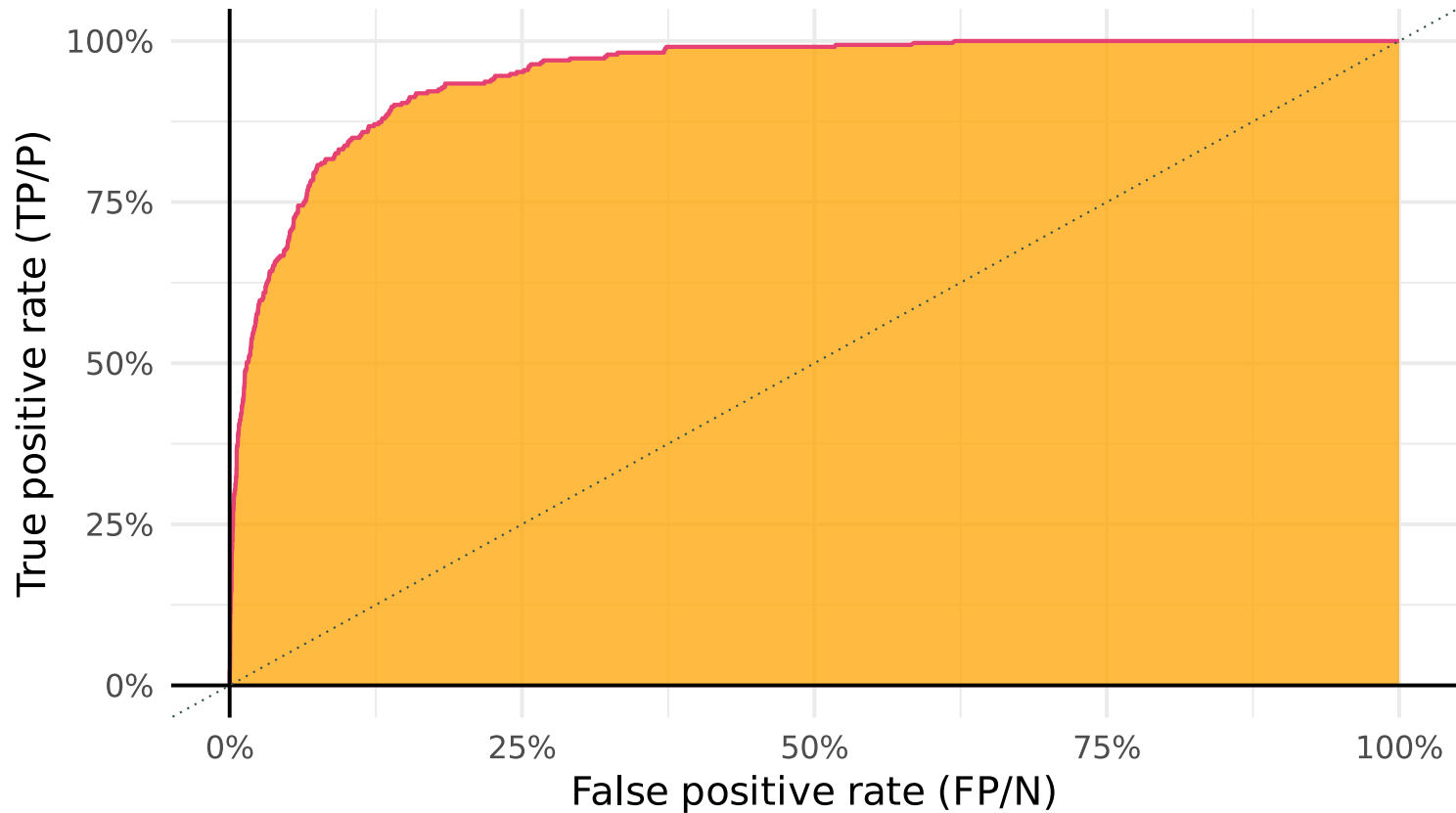


"Best performance" means the **ROC curve** hugs the top-left corner.

The **AUC** gives the area under the (ROC) curve.



The **AUC** gives the area under the (ROC) curve.



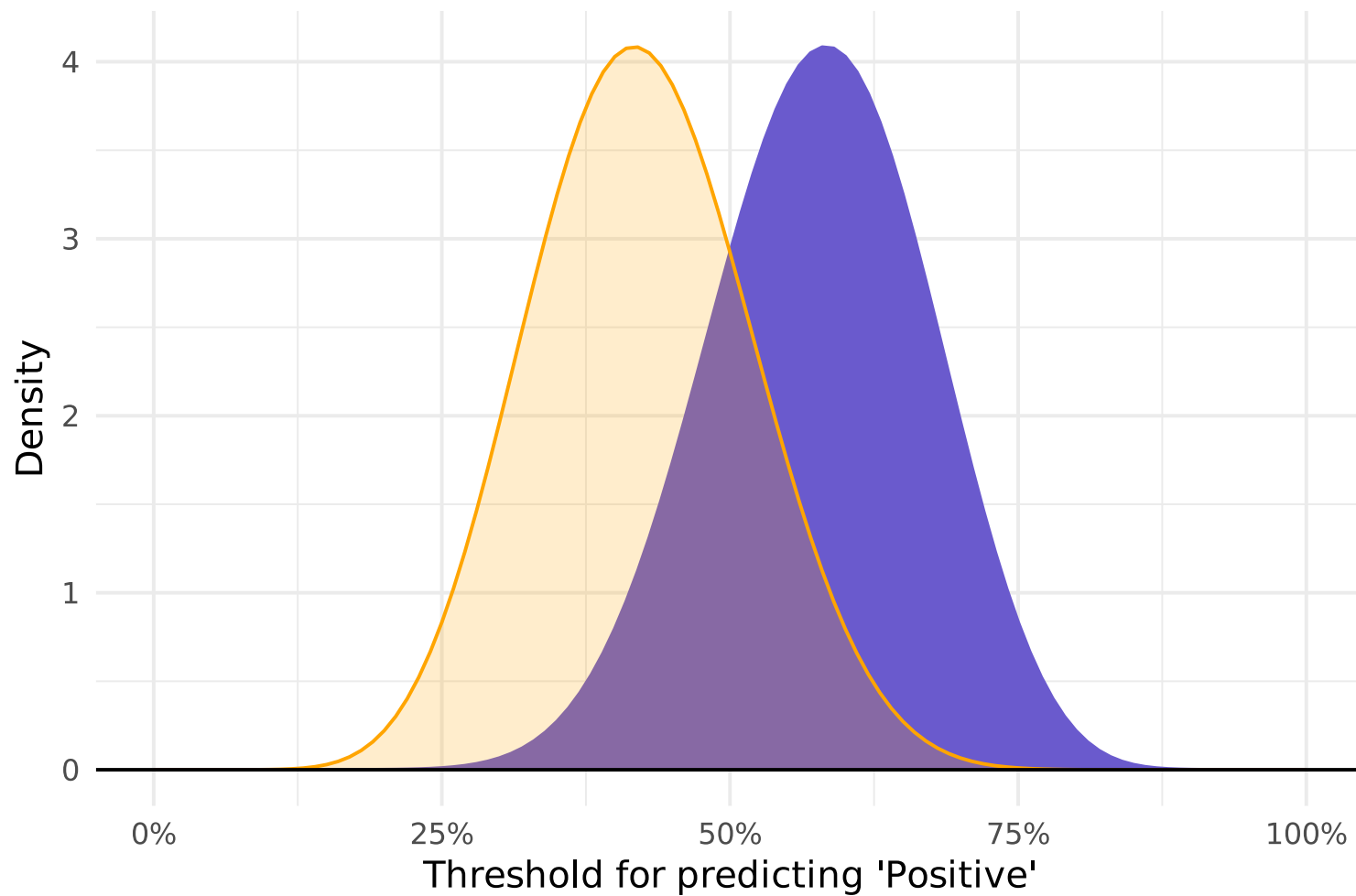
"Best performance" means the **AUC** is near 1. Random chance: 0.5

Q So what information is AUC telling us?

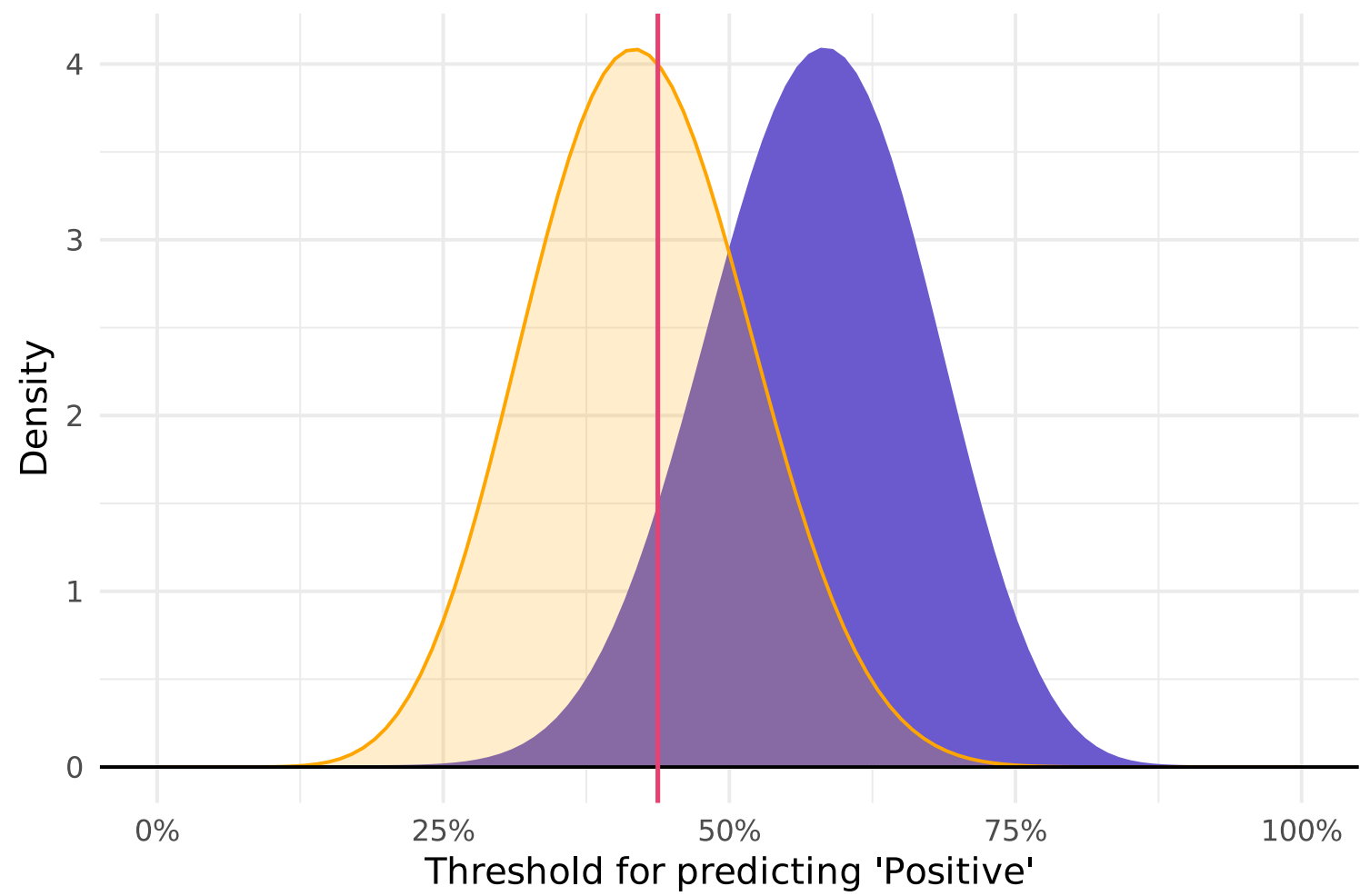
Q So what information is AUC telling us?

A AUC tells us how much we've **separated** the *positive* and *negative* labels.

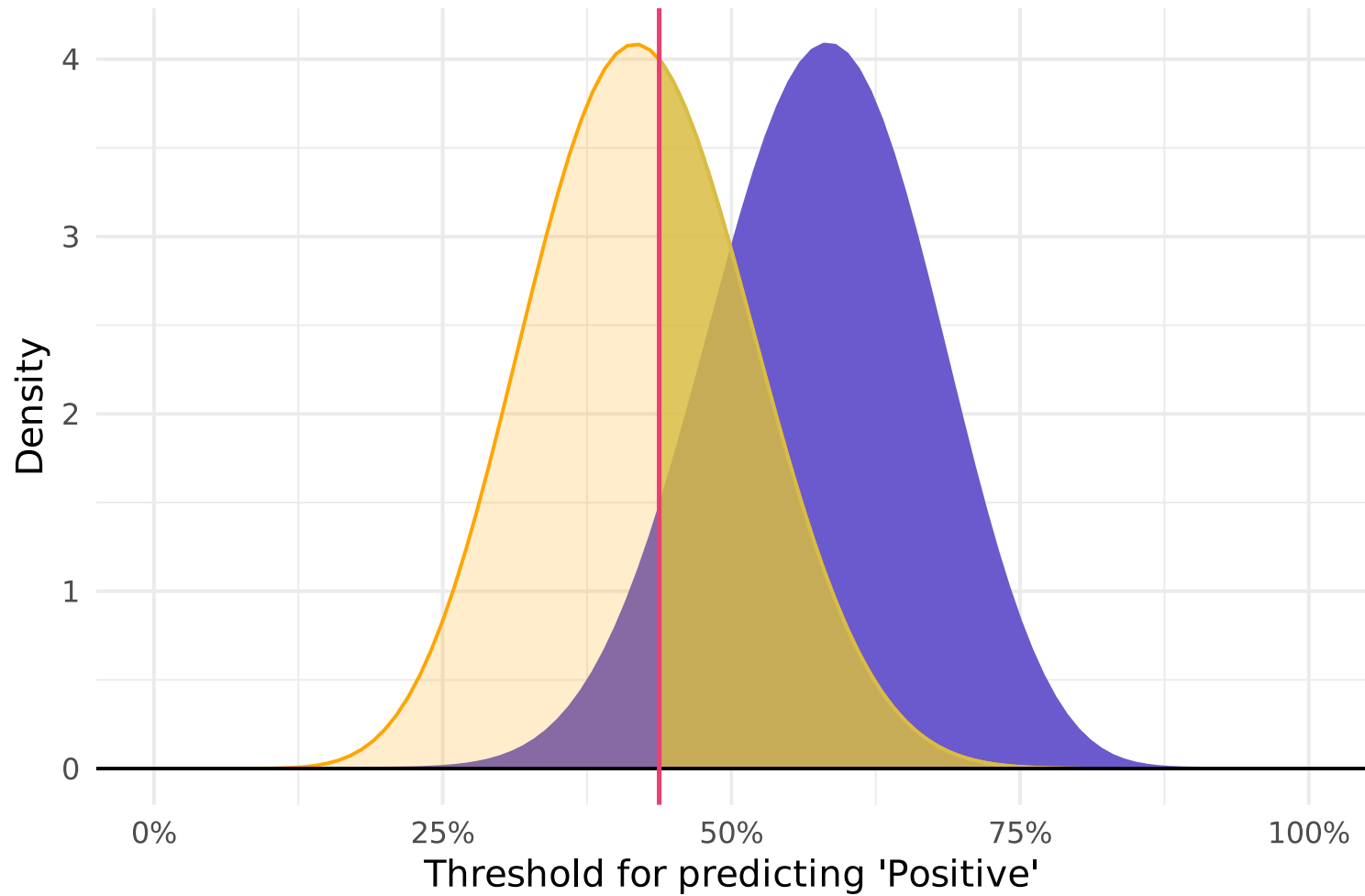
Example: Distributions of probabilities for **negative** and **positive** outcomes.



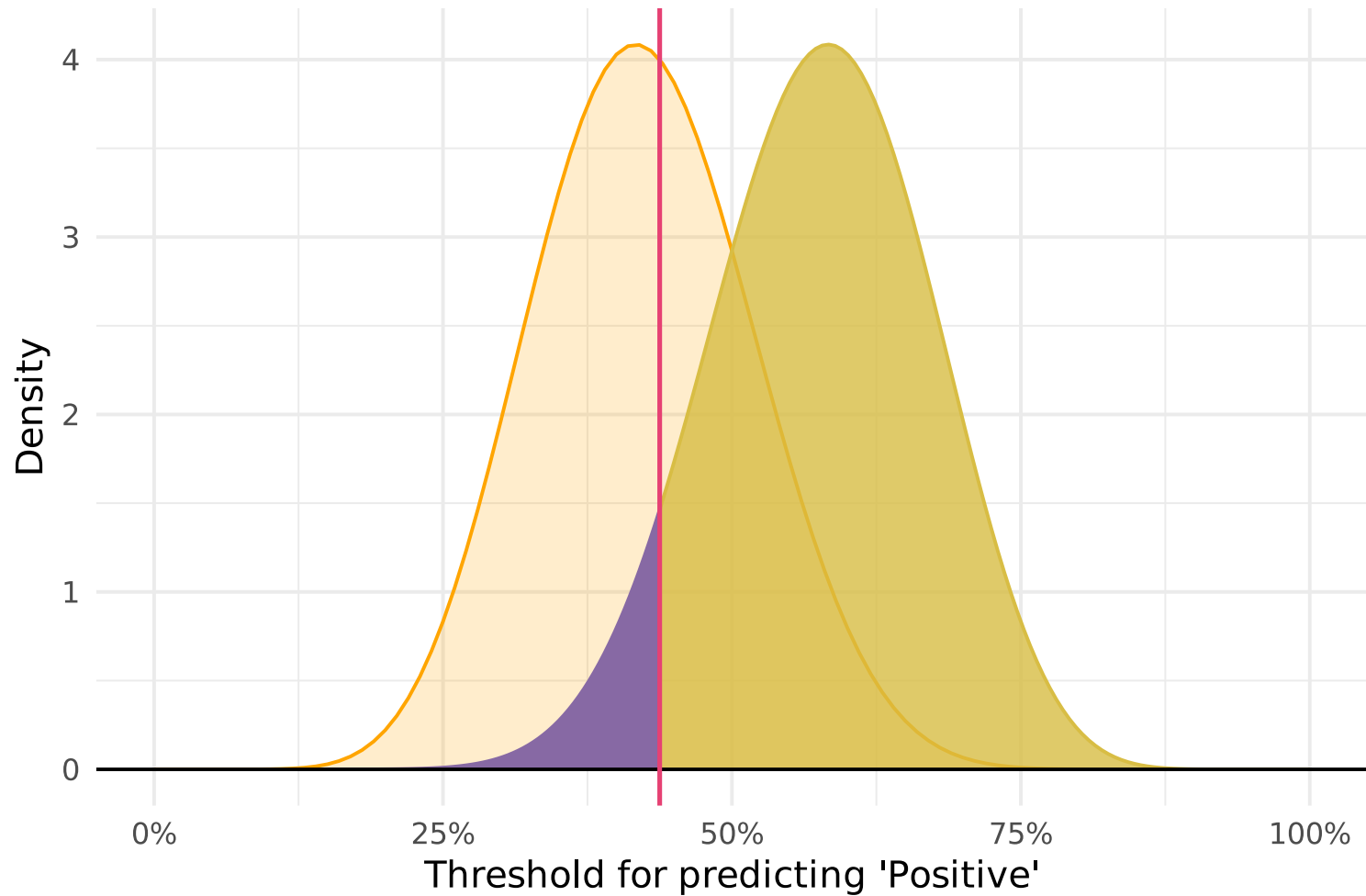
For any given **threshold**



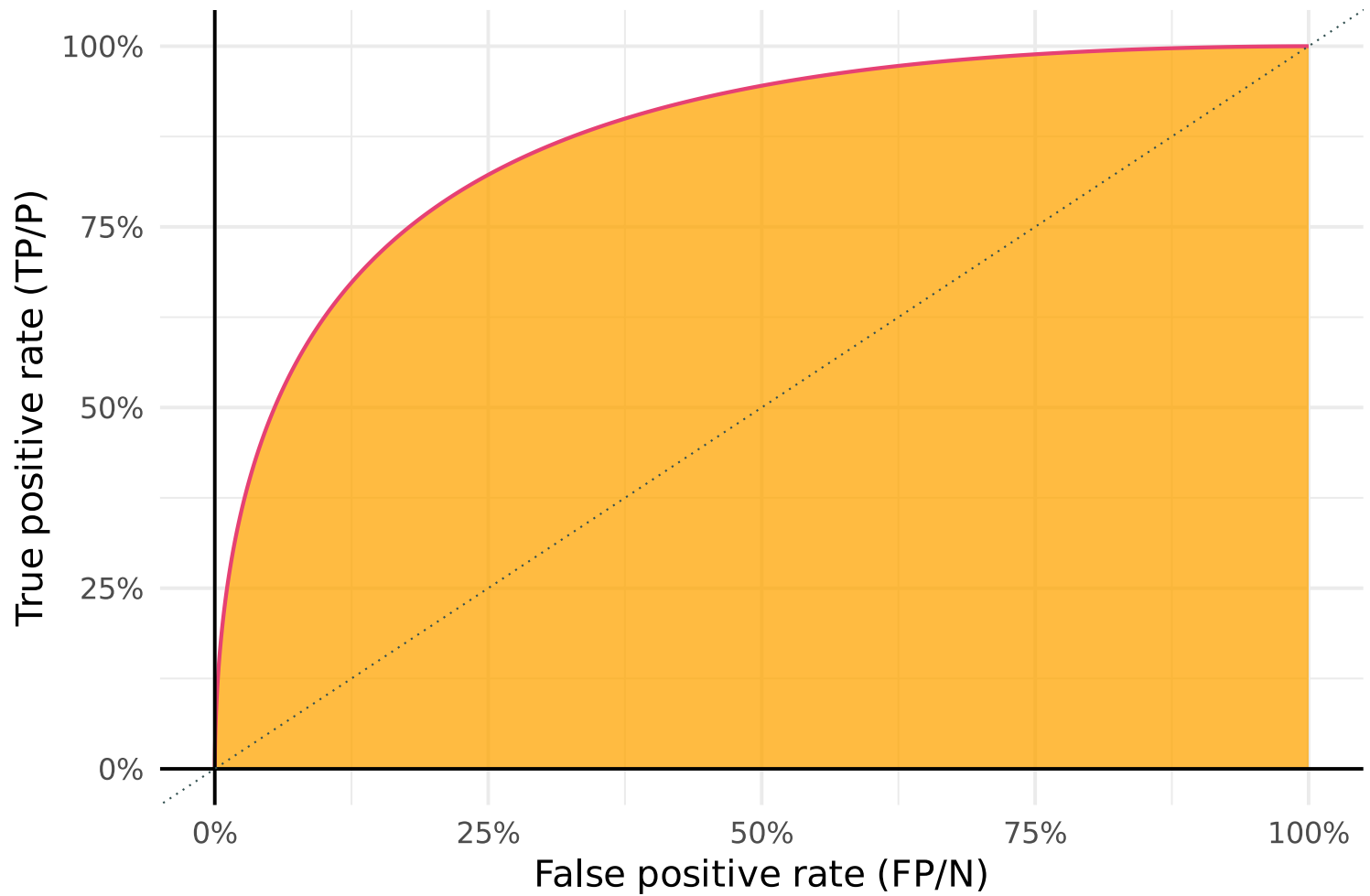
For any given **threshold**, we get **false positives**



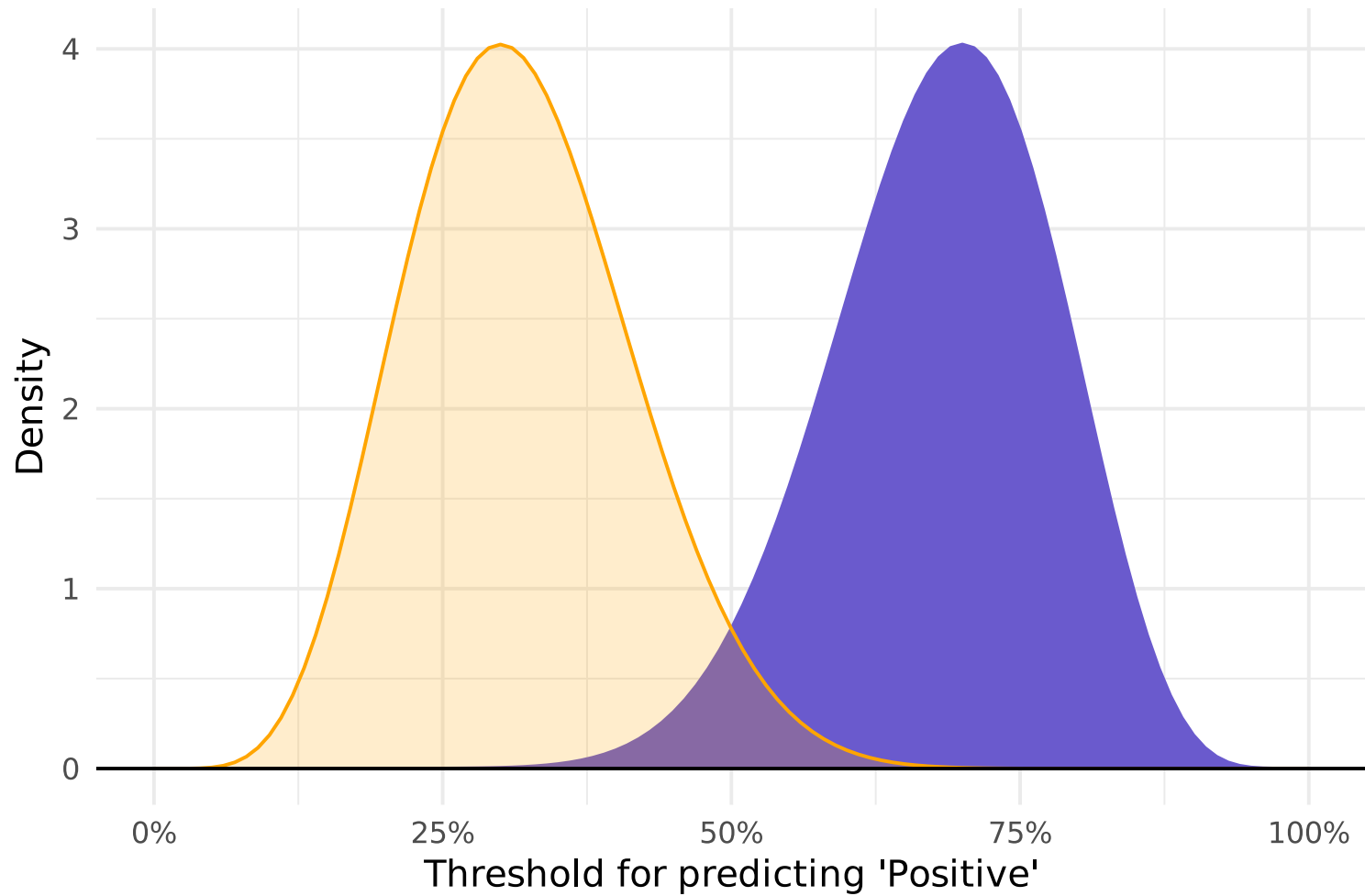
For any given **threshold**, we get false positives and **true positives**.



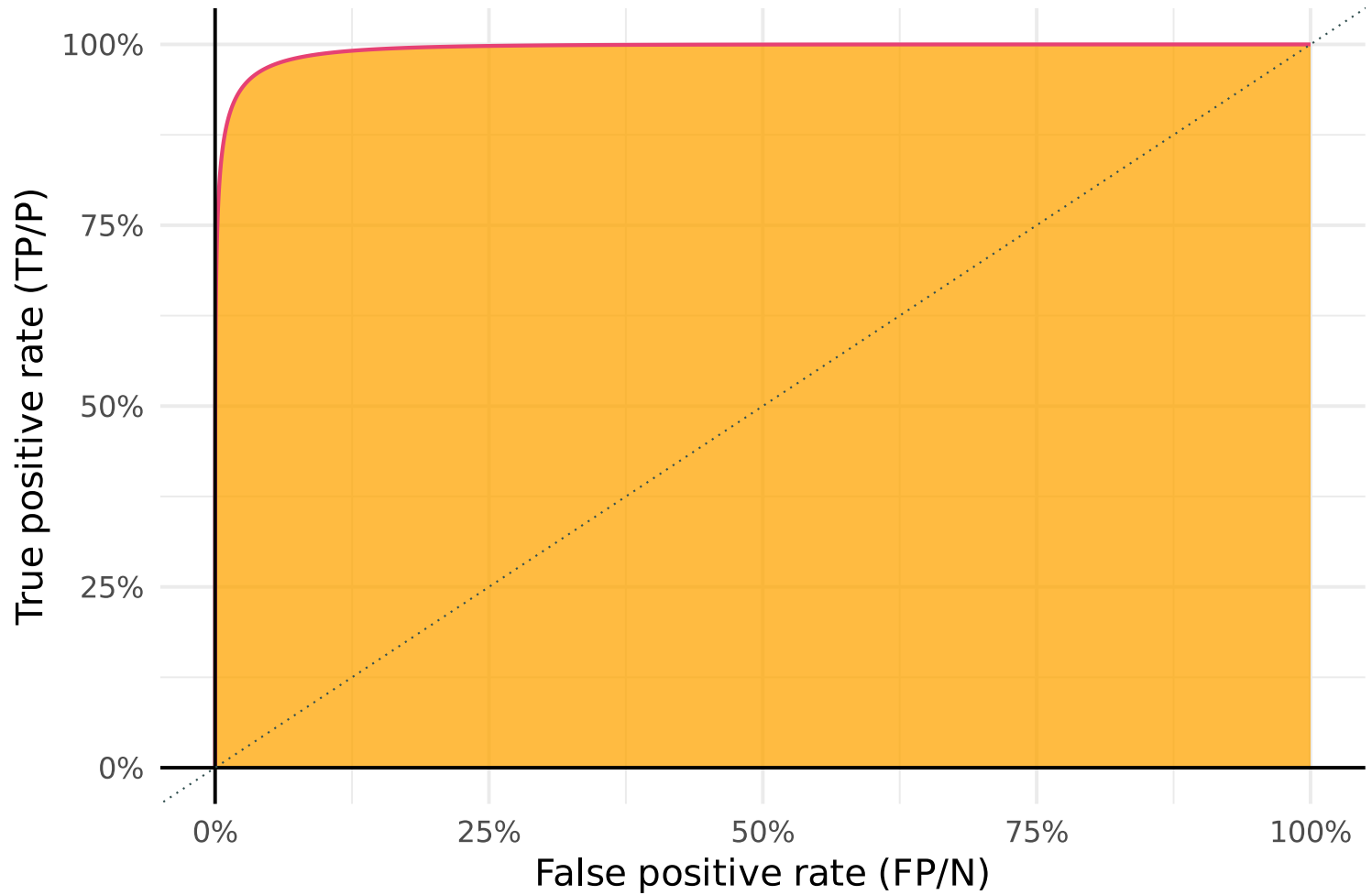
... moving through all possible thresholds generates the **ROC** (**AUC** ≈ 0.872).



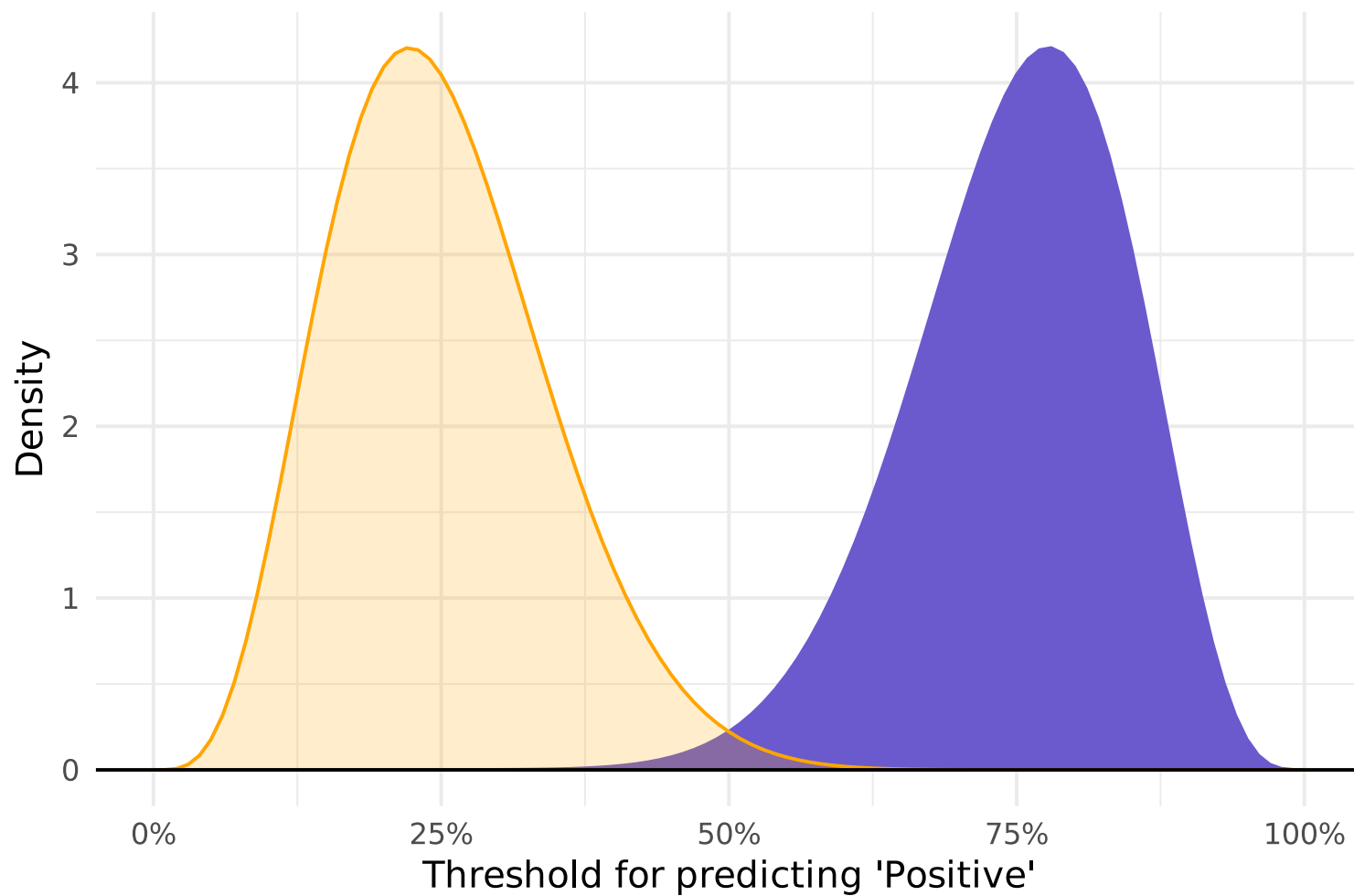
Increasing separation between **negative** and **positive** outcomes...



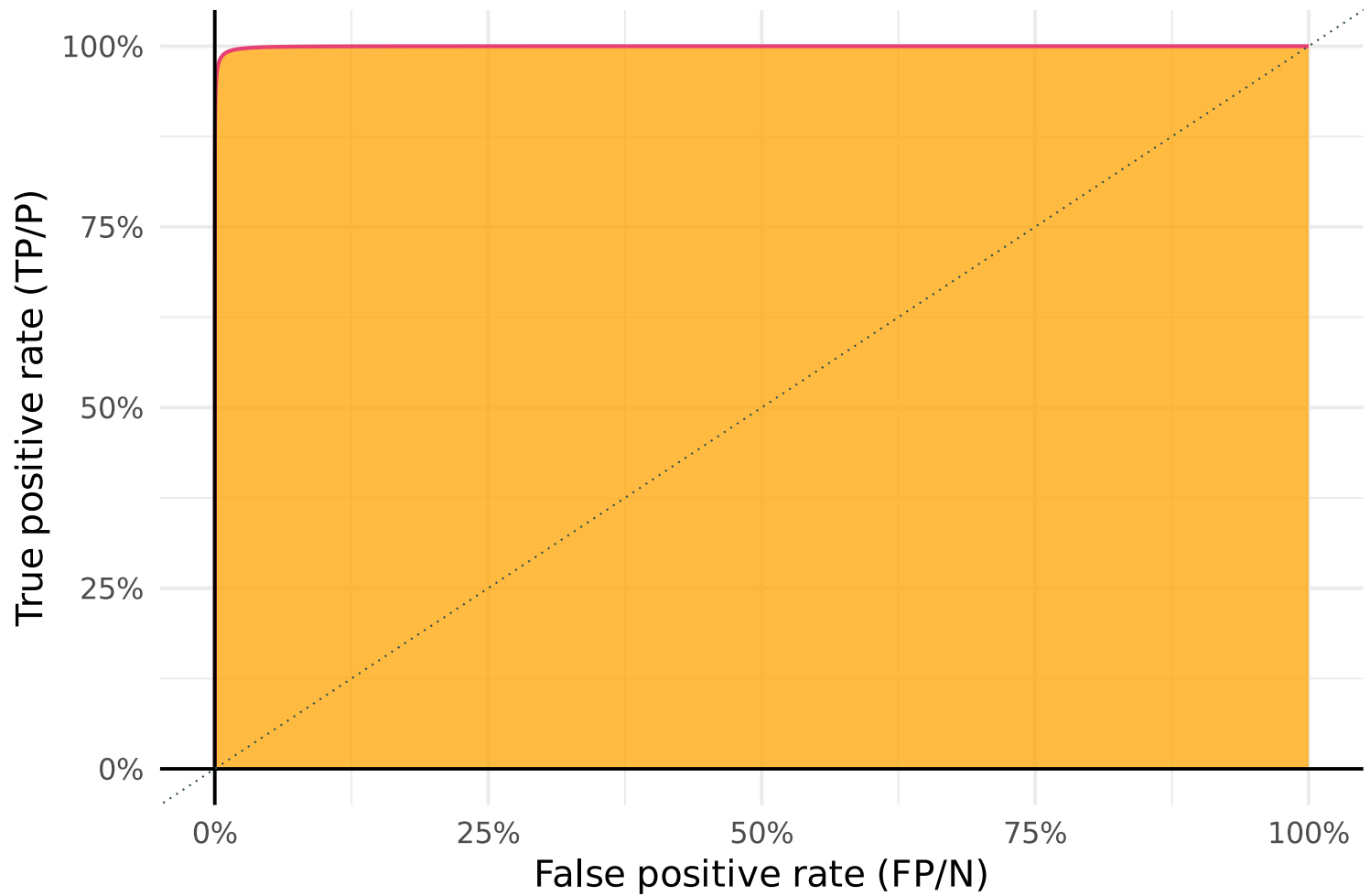
... reduces error (shifts **ROC**) and increases **AUC** (≈ 0.994).



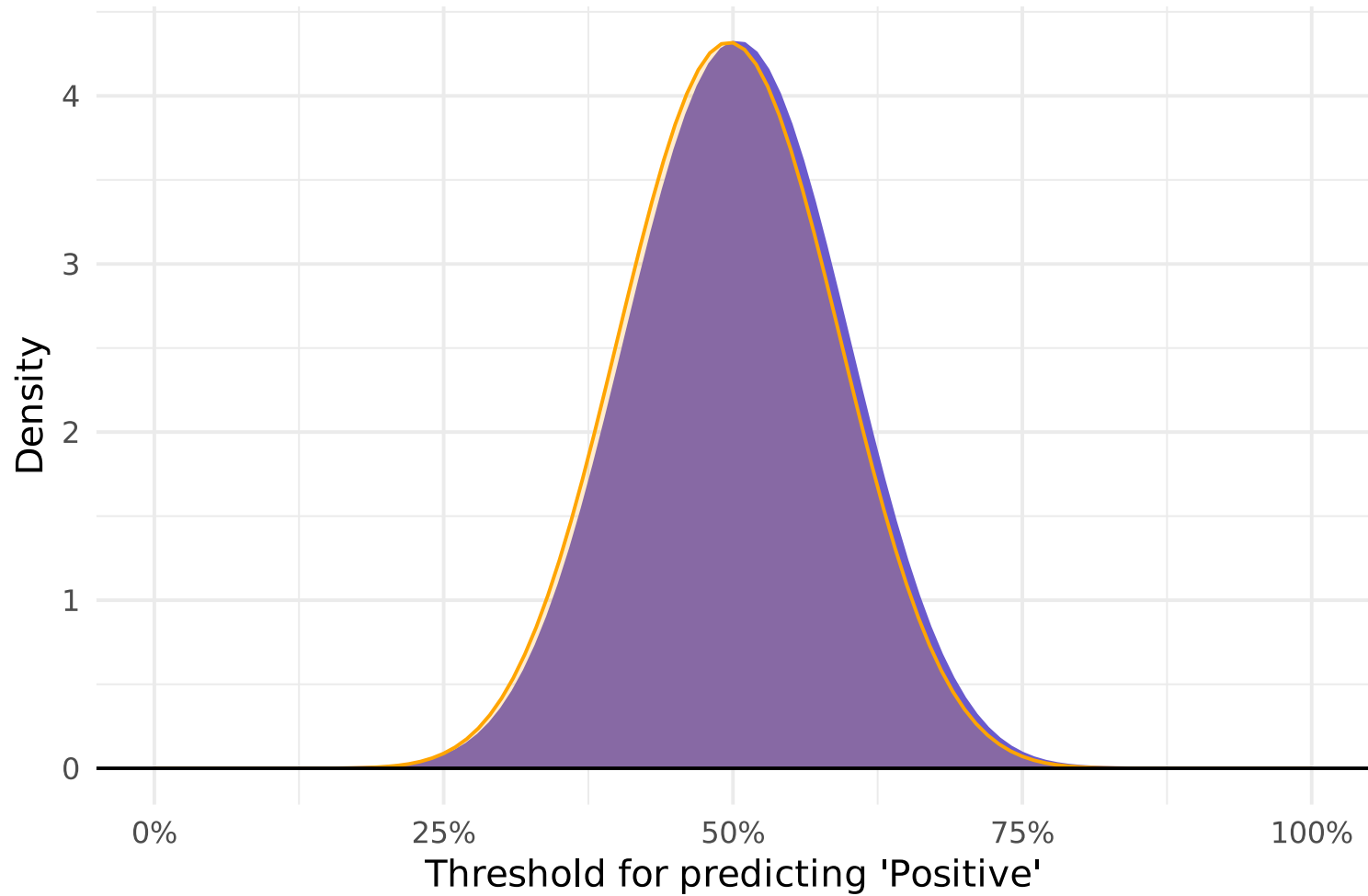
Further increasing separation between **negative** and **positive** outcomes...



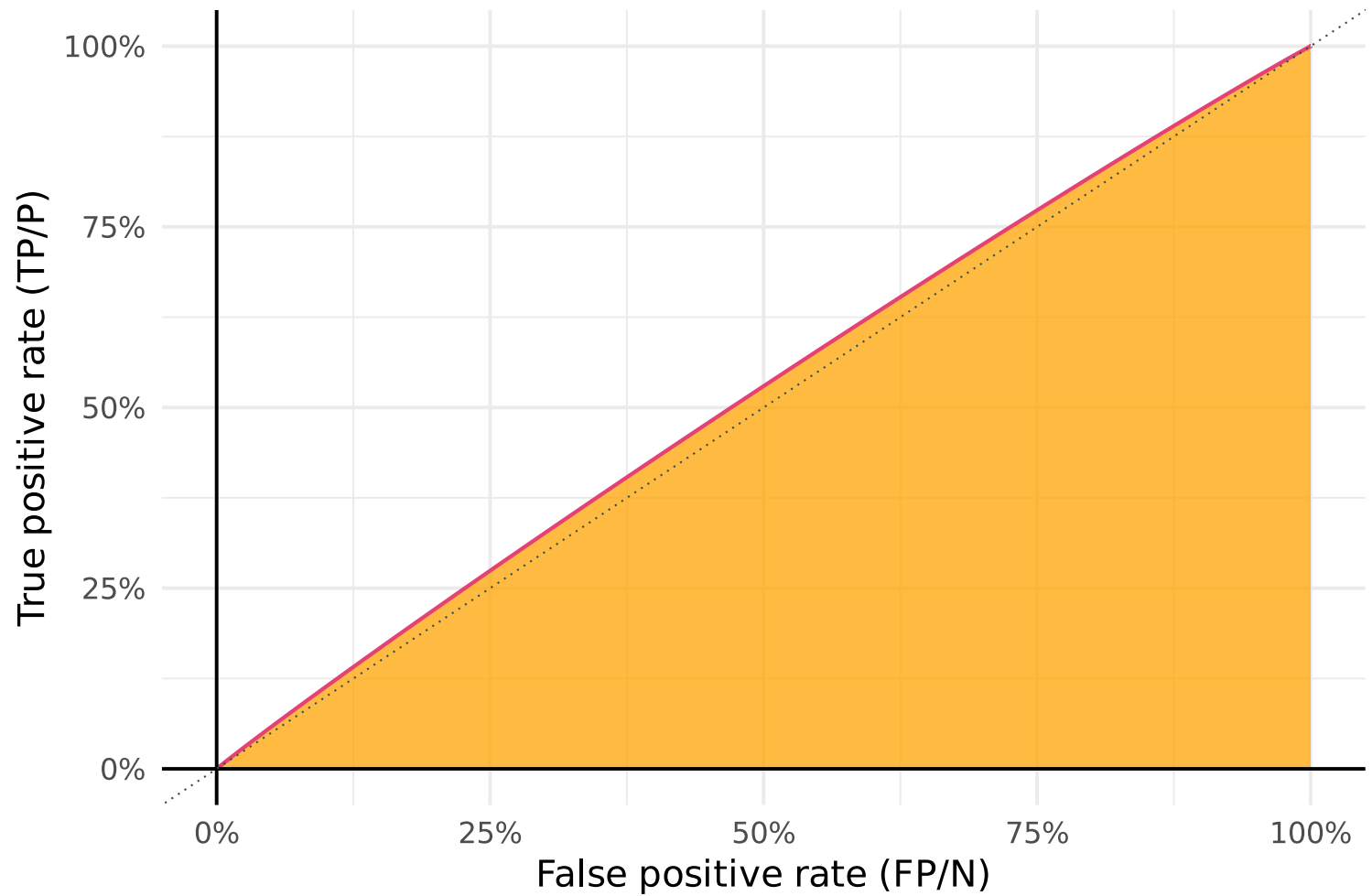
... reduces error (shifts **ROC**) and increases **AUC** (≈ 1).



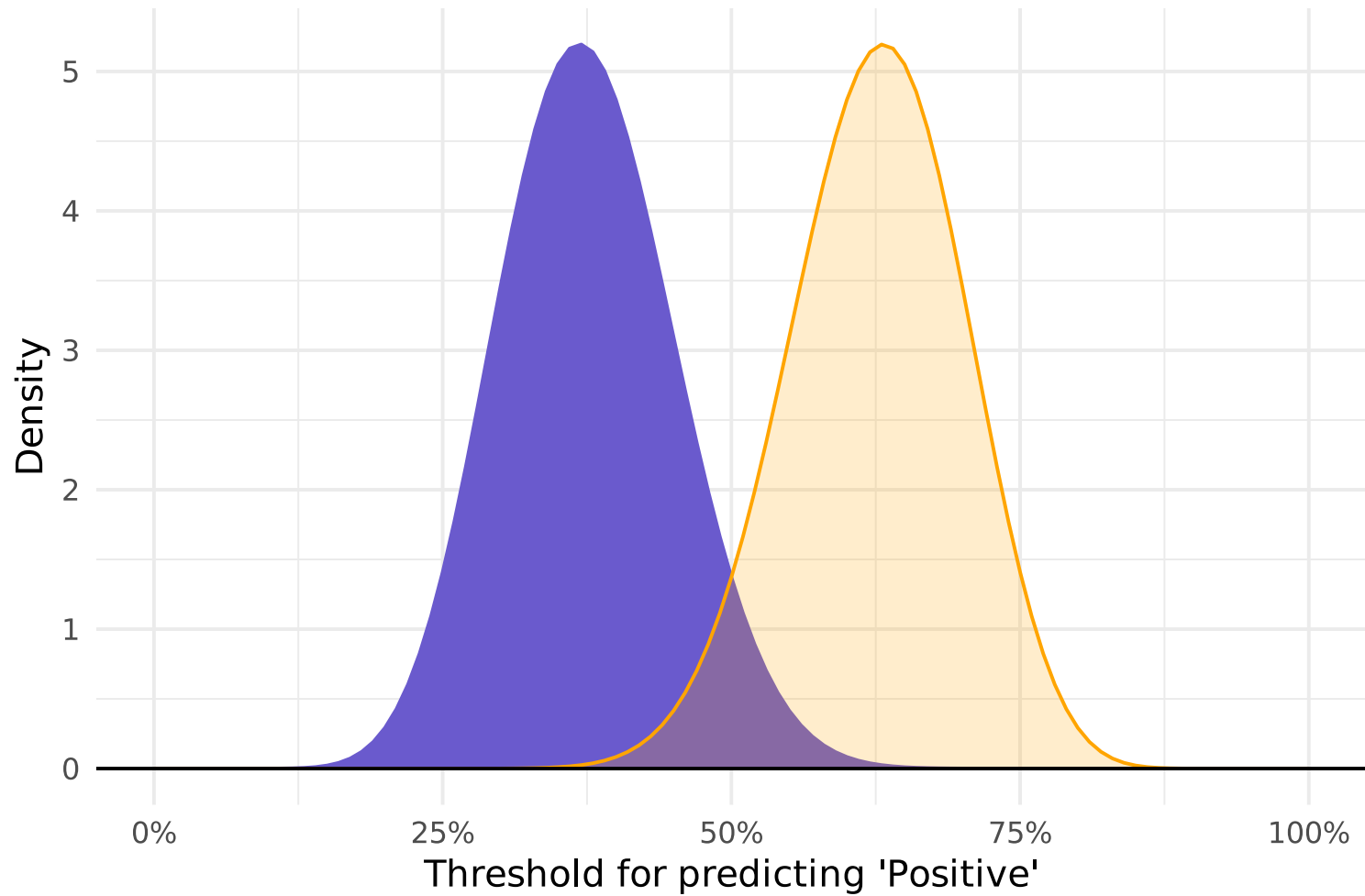
Tiny separation ("guessing") between **negative** and **positive** outcomes...



... increases error (shifts **ROC**) and pushes **AUC** toward 0.5 (here ≈ 0.523).



Getting **negative** and **positive** outcomes backwards...



... increases error (shifts **ROC**) and pushes **AUC** toward 0 (here ≈ 0.012).



R extras

AUC You can calculate AUC in R using the `roc_auc()` function from `yardstick`. See the documentation for examples.

Logistic elasticnet `glmnet()` (for ridge, lasso, and elasticnet) extends to logistic regression[†] by specifying the `family` argument of `glmnet`, e.g.,

```
# Example of logistic regression with lasso
logistic_lasso = glmnet(
  y = y,
  x = x,
  family = "binomial",
  alpha = 1,
  lambda = best_lambda
)
```

You can also use the `"glmnet"` engine for `logistic_reg()` in `parsnip`.

[†] Or many other generalized linear models.

Sources

These notes draw upon

- *An Introduction to Statistical Learning (ISL)*
James, Witten, Hastie, and Tibshirani
- *Receiver Operating Characteristic Curves Demystified (in Python)*

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